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United States Patent Application Publication

20250286511

Kind Code

A1

Publication Date

September 11, 2025

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TRACKER CIRCUIT, COMMUNICATION DEVICE, AND TRACKING METHOD

Abstract

A tracker circuit is provided that includes a switched-capacitor circuit configured to generate a plurality of discrete voltages and a supply modulator configured to selectively output at least one discrete voltage of the plurality of discrete voltages to a power amplifier. The switched-capacitor circuit generates the plurality of discrete voltages based on a difference between a first input voltage and a second input voltage that is variable and that is lower than the first input voltage.

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Family ID: 91717636

Appl. No.: 19/215743

Filed: May 22, 2025

Related U.S. Application Data

parent WO continuation PCT/JP2023/044498 20231212 PENDING child US 19215743
us-provisional-application US 63477420 20221228

Publication Classification

Int. Cl.: H03F1/02 (20060101); H03F3/24 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/JP2023/044498, filed Dec. 12, 2023, which claims priority to U.S. Provisional Patent Application No. 63/477,420, filed Dec. 28, 2022, the entire contents of each of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The exemplary aspects of the present disclosure relates to a tracker circuit, a communication device, and a tracking method.

BACKGROUND

[0003] In general, U.S. Pat. No. 8,829,993 describes a digital envelope tracking (ET) mode of supplying a plurality of discrete voltages to a power amplifier leads to improvement in power efficiency.

[0004] However, further improvement in power efficiency is desired in the digital ET mode, such as the existing systems and methods described in U.S. Pat. No. 8,829,993.

SUMMARY OF THE INVENTION

[0005] Accordingly, the exemplary aspects of the present disclosure provide a tracker circuit, a communication device, and a tracking method that are enabled to improve power efficiency.

[0006] In an exemplary aspect, a tracker circuit is provided that includes a switched-capacitor circuit configured to generate a plurality of discrete voltages; and a supply modulator configured to selectively output at least one discrete voltage of the plurality of discrete voltages to a power amplifier. The switched-capacitor circuit is configured to generate the plurality of discrete voltages, based on a voltage difference between a first input voltage and a second input voltage that is variable and that is lower than the first input voltage.

[0007] In another exemplary aspect, a tracker circuit is provided that includes a switched-capacitor circuit that includes a first input terminal and a second input terminal and that is configured to generate a plurality of discrete voltages, the first input terminal configured to receive a first input voltage, the second input terminal configured to receive a second input voltage that is variable and that is lower than the first input voltage; a supply modulator configured to selectively output at least one discrete voltage of the plurality of discrete voltages to a power amplifier; a first capacitor connected between the first input terminal and ground; and a second capacitor connected between the second input terminal and ground.

[0008] In yet another exemplary aspect, a tracking method is provided that includes receiving a first input voltage and a second input voltage lower than the first input voltage; generating a plurality of first discrete voltages based on a first voltage difference between the first input voltage and the second input voltage; selectively outputting at least one discrete voltage of the plurality of first discrete voltages to a power amplifier; receiving a third input voltage and a fourth input voltage that is lower than the third input voltage, the fourth input voltage being different from the second input voltage; generating a plurality of second discrete voltages based on a second voltage difference between the third input voltage and the fourth input voltage; and selectively outputting at least one discrete voltage of the plurality of second discrete voltages to the power amplifier.

[0009] With tracker circuit and the like according to the exemplary aspects of the present disclosure, power efficiency is improved.

Description

BRIEF DESCRIPTION OF DRAWINGS

- [0010] FIG. 1A is a graph illustrating an example of power supply voltage changes in an APT mode.
- [0011] FIG. 1B is a graph illustrating an example of power supply voltage changes in an analog ET mode.
- [0012] FIG. 1C is a graph illustrating an example of power supply voltage changes in a digital ET mode.
- [0013] FIG. 2 is a circuit configuration diagram of a communication device according to Exemplary Embodiment 1.
- [0014] FIG. 3 is a circuit configuration diagram of a tracker circuit and a pre-regulator circuit according to Exemplary Embodiment 1.
- [0015] FIG. 4 is a flowchart illustrating a tracking method according to Exemplary Embodiment 1.
- [0016] FIG. 5A is a graph illustrating a radio frequency signal modulated by QPSK and a plurality of discrete voltages.
- [0017] FIG. 5B is a graph illustrating a relationship between the output power of a power amplifier that amplifies a radio frequency signal modulated by QPSK and an occurrence frequency.
- [0018] FIG. 5C is a graph illustrating a relationship between the output power of the power amplifier that amplifies a radio frequency signal modulated by QPSK and power efficiency.
- [0019] FIG. 6A is a graph illustrating a radio frequency signal modulated by 256QAM and a plurality of discrete voltages.
- [0020] FIG. 6B is a graph illustrating a relationship between the output power of a power amplifier that amplifies a radio frequency signal modulated by 256QAM and an occurrence frequency.
- [0021] FIG. 6C is a graph illustrating a relationship between the output power of the power amplifier that amplifies a radio frequency signal modulated by 256QAM and power efficiency.
- [0022] FIG. 7 is a circuit configuration diagram of a tracker circuit according to a modification of Exemplary Embodiment 1.
- [0023] FIG. 8 is a communication device according to Exemplary Embodiment 2.
- [0024] FIG. 9 is a circuit configuration diagram of a tracker circuit and pre-regulator circuits according to Exemplary Embodiment 2.
- [0025] FIG. 10 is a circuit configuration diagram of a communication device according to Exemplary Embodiment 3.
- [0026] FIG. 11 is a circuit configuration diagram of a tracker circuit and a pre-regulator circuit according to Exemplary Embodiment 3.

DETAILED DESCRIPTION OF EMBODIMENTS

[0027] Hereinafter, exemplary embodiments of the present disclosure will be described in detail by using the drawings. The embodiments described below each illustrate a general or specific example. The numerical values, shapes, materials, components, the disposition and connection form of the components, and the like described in the following embodiments are examples and are not intended to limit the present disclosure.

[0028] It is noted that the drawings are schematic diagrams having undergone emphasis, omission, or ratio adjustment performed as appropriate to illustrate the exemplary aspects of the present disclosure. The illustration therein is not necessarily strict and is different from actual shapes, positional relationships, and ratios on occasions. In the drawings, components that are substantially the same are denoted by the same reference numerals, and a repeated description thereof is omitted or simplified in some cases.

[0029] In the circuit configurations of the exemplary aspects of the present disclosure, the term “connected” can refer to not only a direct connection using a connection terminal and/or a wiring

conductor but also an electrical connection via another circuit element. Moreover, the phrase “directly connected” can mean directly connected using a connection terminal and/or a wiring conductor without interposing another circuit element. The phrase “C is connected between A and B” can mean that one end of C is connected to A and the other end of C is connected to B. Moreover, the phrase “a path between A and B” can mean a path composed of a conductor that electrically connects A to B.

[0030] In the circuit configurations of the exemplary aspects of the present disclosure, the term “terminal” can refer to a point at which a conductor in an element terminates. In a case where the impedance of a conductor between elements is sufficiently low, a terminal is interpreted as not only a single point but also any point on the conductor between the elements or the entire conductor.

[0031] In addition, terms indicating relationships between elements, such as “parallel” and “perpendicular”, terms indicating the shapes of elements, such as “rectangular”, and numerical ranges do not represent only strict meanings, but include substantially equivalent ranges, for example, an error of about several percent.

[0032] A description will be given below of tracking modes in which a power amplifier is supplied with a power supply voltage dynamically regulated based on a radio frequency signal over time, which are techniques for amplifying the radio frequency signal with high efficiency. According to an exemplary aspect, a tracking mode is a mode of dynamically regulating the power supply voltage to be applied to a power amplifier. Among several types of tracking modes, a description will herein be given of an average power tracking (APT) mode, an analog ET mode, and a digital ET mode with reference to FIG. 1A to FIG. 1C. In FIG. 1A to FIG. 1C, the horizontal axis represents time, and the vertical axis represents voltage. A thick solid line represents a power supply voltage to be supplied to a power amplifier, and a thin solid line (e.g., waveform) represents a modulated signal.

[0033] FIG. 1A is a graph illustrating an example of power supply voltage changes in the APT mode. In the APT mode, the power supply voltage is varied to a plurality of discrete voltage levels in units of one frame based on average power. As the result, a power supply voltage signal forms a rectangular wave.

[0034] According to an exemplary aspect, a frame can refer to a unit of elements of a radio frequency signal (e.g., a modulated signal). For example, in 5th Generation New Radio (5G NR) and Long Term Evolution (LTE), a frame includes ten subframes, each subframe includes a plurality of slots, and each slot is composed of a plurality of symbols. The subframe has a length of 1 ms, and the frame has a length of 10 ms according to an exemplary aspect.

[0035] For purposes of this disclosure, a mode of varying a voltage level in units of one frame or in larger units based on average power is referred to as an APT mode and is distinguished from a mode of varying a voltage level in units smaller than one frame (for example, in units of subframes, slots, or symbols). For example, a mode of varying a voltage level in units of symbols is referred to as a symbol power tracking (SPT) mode and is distinguished from the APT mode.

[0036] FIG. 1B is a graph illustrating an example of power supply voltage changes in the analog ET mode. In the analog ET mode, the power supply voltage is continuously varied based on an envelope signal, and thereby the envelope of a modulated signal is tracked.

[0037] The envelope signal is a signal indicating the envelope of a modulated signal. An envelope value is represented by, for example, the square root of $(I_{\text{sup}}^2 + Q_{\text{sup}}^2)$. It is noted that (I, Q) can represent a constellation point herein. The constellation point is a point representing, on a constellation diagram, a signal modulated by digital modulation. A point (I, Q) is decided, for example, by a baseband integrated circuit (BBIC), for example, based on sending information.

[0038] FIG. 1C is a graph illustrating an example of power supply voltage changes in the digital ET mode. In the digital ET mode, the power supply voltage is varied to a plurality of discrete voltage levels within one frame based on an envelope signal, and thereby the envelope of a modulated signal is tracked. As the result, the power supply voltage signal forms a rectangular wave.

Exemplary Embodiment 1

[0039] Hereinafter, Exemplary Embodiment 1 will be described.

[1.1 Circuit Configuration of Communication Device 7]

[0040] First, a communication device 7 according to exemplary embodiment 1 will be described with reference to FIG. 2. FIG. 2 is a circuit configuration diagram of the communication device 7 according to exemplary embodiment 1.

[0041] FIG. 2 illustrates an exemplary circuit configuration, and the communication device 7 may be implemented by using any of a wide variety of circuit implementation and circuit techniques. Accordingly, the description of the communication device 7 provided below should not be construed in a limiting manner.

[0042] In the exemplary aspect, the communication device 7 corresponds to a user terminal (user equipment (UE)) in a cellular network and is typically a mobile phone, a smartphone, a tablet computer, a wearable device, or the like. The communication device 7 may be an Internet of Things (IoT) sensor device, a medical/health-care device, a vehicle, an unmanned aerial vehicle (UAV) (a so-called drone), or an automated guided vehicle (AGV). The communication device 7 can be configured to function as a base station (BS) in a cellular network.

[0043] As illustrated in FIG. 2, the communication device 7 includes a tracker circuit 1, a radio frequency circuit 4, a radio frequency integrated circuit (RFIC) 5, an antenna 6, a pre-regulator circuit 10, and a direct current power source 50.

[0044] According to an exemplary aspect, the tracker circuit 1 is configured to supply a plurality of discrete voltages to a power amplifier 2 in the digital ET mode. As illustrated in FIG. 2, the tracker circuit 1 includes a switched-capacitor circuit 20, a supply modulator 30, and a digital control circuit 60.

[0045] The switched-capacitor circuit 20 is a differential switched-capacitor circuit and is also referred to as a differential switched-capacitor voltage balancer on occasions. According to an exemplary aspect, the switched-capacitor circuit 20 is configured to generate a plurality of discrete voltages including a first output voltage, a second output voltage, and a third output voltage, based on a difference between a first input voltage and a second input voltage received from the pre-regulator circuit 10. Specifically, the switched-capacitor circuit 20 is configured to generate the third output voltage to cause each of a difference between the first output voltage and the third output voltage and a difference between the second output voltage and the third output voltage to increase in response to an increase in the voltage difference between the first input voltage and the second input voltage. It is thus possible to control the step widths of the plurality of discrete voltages by using the voltage difference between the first input voltage and the second input voltage. In this case, the first output voltage is based on the first input voltage and is equal to the first input voltage in this embodiment. The second output voltage is based on the second input voltage and is equal to the second input voltage in this embodiment. The third output voltage is based on the voltage difference between the first input voltage and the second input voltage and is lower than the first output voltage (first input voltage) and higher than the second output voltage (second input voltage) in this embodiment. The circuit configuration of the switched-capacitor circuit 20 will be described later by using FIG. 3.

[0046] According to an exemplary aspect, the supply modulator 30 is configured to selectively output, to the power amplifier 2, at least one of the plurality of discrete voltages generated by the switched-capacitor circuit 20. The supply modulator 30 is thus configured to select at least one voltage from among the plurality of discrete voltages and outputting the selected voltage to the power amplifier 2. The circuit configuration of the supply modulator 30 will be described later by using FIG. 3.

[0047] According to an exemplary aspect, the digital control circuit 60 is configured to control the switched-capacitor circuit 20 and the supply modulator 30 based on digital control signals from the RFIC 5. Specifically, the digital control circuit 60 is configured to generate and outputting a control

signal for controlling a switch included in the switched-capacitor circuit **20** and a control signal for controlling a switch included in the supply modulator **30**. The circuit configuration of the digital control circuit **60** will be described later by using FIG. **3**. It should be appreciated that the digital control circuit **60** may be omitted from the tracker circuit **1** in an exemplary aspect.

[0048] The pre-regulator circuit **10** is an example of a converter circuit and is also referred to as a magnetic regulator or a direct current (DC)/DC converter on occasions. In this embodiment, the pre-regulator circuit **10** is a one-input two-output buck-boost converter and is configured to convert a power supply voltage supplied from the direct current power source **50** into two variable voltages independent of each other (the first input voltage and the second input voltage of the switched-capacitor circuit **20**). The pre-regulator circuit **10** is configured to change the first input voltage and the second input voltage of the switched-capacitor circuit **20** based on, for example, a control signal from the RFIC **5**. Specifically, the pre-regulator circuit **10** is configured to switch the voltage difference between the first input voltage and the second input voltage in accordance with, for example, a modulation scheme for a radio frequency signal amplified by the power amplifier **2**. More specifically, the pre-regulator circuit **10** can be configured to increase the voltage difference between the first input voltage and the second input voltage as the bit rate of the modulation scheme increases. Such switching of the voltage difference between the first input voltage and the second input voltage may be performed at the time of switching of a frame of the radio frequency signal. The circuit configuration of the pre-regulator circuit **10** will be described later by using FIG. **3**.

[0049] The modulation scheme for the radio frequency signal enables constellation points of the radio frequency signal to be identified by measurement. The number of separable constellation points increases as the bit rate of the modulation scheme increases.

[0050] A condition for switching the voltage difference between the first input voltage and the second input voltage is not limited to the modulation scheme. For example, the voltage difference between the first input voltage and the second input voltage may be switched in accordance with a power class or the like.

[0051] According to an exemplary aspect, the direct current power source **50** is configured to supply a power supply voltage to the pre-regulator circuit **10**. The direct current power source **50** may be, but is not limited to, a rechargeable battery, for example.

[0052] According to an exemplary aspect, the radio frequency circuit **4** is configured to transmit a radio frequency signal between the RFIC **5** and the antenna **6**. As illustrated in FIG. **2**, the radio frequency circuit **4** includes the power amplifier **2** and a filter **3**.

[0053] The power amplifier **2** is connected between the RFIC **5** and the filter **3**. The power amplifier **2** is further connected to the tracker circuit **1**. The power amplifier **2** is configured to amplify a radio frequency signal received from the RFIC **5**, by using a plurality of discrete voltages received from the tracker circuit **1**.

[0054] The filter **3** is connected between the power amplifier **2** and the antenna **6**. The filter **3** is a band pass filter having a pass band including a predetermined band. It should be appreciated that the filter **3** may be omitted from the radio frequency circuit **4** in an exemplary aspect.

[0055] The predetermined band is a frequency band for a communication system constructed by using radio access technology (RAT) and is predefined by standardizing bodies (such as 3rd Generation Partnership Project (3GPP®) and Institute of Electrical and Electronics Engineers (IEEE)). As examples of the communication system, a 5G NR system, an LTE system, a Wireless Local Area Network (WLAN) system, and the like can be cited.

[0056] The antenna **6** sends a radio frequency signal received from the radio frequency circuit **4**. It should be appreciated that the antenna **6** may be omitted from the communication device **7** in an exemplary aspect.

[0057] The circuit configuration of the communication device **7** illustrated in FIG. **2** is illustrative and is not restrictive. For example, the communication device **7** may include a baseband signal

processing circuit that performs signal processing by using an intermediate frequency band lower than the frequency of a radio frequency signal.

[1.2 Circuit Configuration of Tracker Circuit 1]

[0058] The circuit configuration of the tracker circuit **1** will then be described with reference to FIG. **3**. FIG. **3** is a circuit configuration diagram of the tracker circuit **1** and the pre-regulator circuit **10** according to this embodiment.

[0059] FIG. **3** illustrates an exemplary circuit configuration, and the tracker circuit **1** and the pre-regulator circuit **10** may be implemented by using any of a wide variety of circuit implementation and circuit techniques. Accordingly, the description of the individual circuits provided below should not be construed in a limiting manner.

[0060] The tracker circuit **1** includes the switched-capacitor circuit **20**, the supply modulator **30**, and the digital control circuit **60**, as described above. The tracker circuit **1** may include a filter circuit (not illustrated) between the supply modulator **30** and the power amplifier **2**.

[0061] Hereinafter, the circuit configurations of the switched-capacitor circuit **20**, the supply modulator **30**, and the digital control circuit **60** will be described in order.

[1.2.1 Circuit Configuration of Switched-Capacitor Circuit 20]

[0062] First, the circuit configuration of the switched-capacitor circuit **20** will be described with reference to FIG. **3**. The switched-capacitor circuit **20** includes capacitors **C11**, **C12**, **C20**, and **C30**, switches **S11** to **S14** and **S21** to **S24**, input terminals **121** and **122**, and output terminals **123** to **125**. Energy and electric charges are input from the pre-regulator circuit **10** to nodes **N1** and **N3** via the input terminals **121** and **122** and output from the nodes **N1** to **N3** to the supply modulator **30** via the output terminals **123** to **125**.

[0063] The input terminal **121** is an example of a first input terminal and is a terminal for receiving a first input voltage (**V3**) from the pre-regulator circuit **10**. The input terminal **121** is connected to the pre-regulator circuit **10** outside the switched-capacitor circuit **20** and is connected to the node **N3** inside the switched-capacitor circuit **20**.

[0064] The input terminal **122** is an example of a second input terminal and is a terminal for receiving a second input voltage (**V1**) from the pre-regulator circuit **10**. The input terminal **122** is connected to the pre-regulator circuit **10** outside the switched-capacitor circuit **20** and is connected to the node **N1** inside the switched-capacitor circuit **20**.

[0065] The output terminal **123** is a terminal for supplying a first output voltage (**V3**) to the supply modulator **30**. The output terminal **123** is connected to the supply modulator **30** outside the switched-capacitor circuit **20** and is connected to the node **N3** inside the switched-capacitor circuit **20**. The output terminal **123** may be integrated with the input terminal **121**.

[0066] The output terminal **124** is a terminal for supplying a third output voltage (**V2**) to the supply modulator **30**. The output terminal **124** is connected to the supply modulator **30** outside the switched-capacitor circuit **20** and is connected to the node **N2** inside the switched-capacitor circuit **20**.

[0067] The output terminal **125** is a terminal for supplying a second output voltage (**V1**) to the supply modulator **30**. The output terminal **125** is connected to the supply modulator **30** outside the switched-capacitor circuit **20** and is connected to the node **N1** inside the switched-capacitor circuit **20**. The output terminal **125** may be integrated with the input terminal **122**.

[0068] The capacitors **C11** and **C12** each function as a flying capacitor (also referred to as a transfer capacitor). Specifically, the capacitors **C11** and **C12** are each used to raise and/or lower the first input voltage and the second input voltage that are supplied from the pre-regulator circuit **10**. More specifically, the capacitors **C11** and **C12** cause electric charges to move between the capacitors **C11** and **C12** and the nodes **N1** to **N3** so that voltages **V1** to **V3** satisfying (**V3**–**V2**): (**V2**–**V1**)=1:1 and **V3**>**V2**>**V1** are maintained at the three nodes **N1** to **N3**.

[0069] The first output voltage (**V3**) is based on the first input voltage and is equal to the first input voltage (**V3**) in this embodiment. The first output voltage is higher than the second output voltage

and the third output voltage.

[0070] The second output voltage (V1) is based on the second input voltage and is equal to the second input voltage (V1) in this embodiment. The second output voltage is lower than the first output voltage and the third output voltage.

[0071] The third output voltage (V2) is based on the voltage difference between the first input voltage (V3) and the second input voltage (V1) and is lower than the first output voltage and higher than the second output voltage in this embodiment.

[0072] The capacitor C11 has two electrodes. One of the two electrodes of the capacitor C11 is connected to one end of the switch S11 and one end of the switch S12. The other of the two electrodes of the capacitor C11 is connected to one end of the switch S21 and one end of the switch S22.

[0073] The capacitor C12 has two electrodes. One of the two electrodes of the capacitor C12 is connected to one end of the switch S13 and one end of the switch S14. The other of the two electrodes of the capacitor C12 is connected to one end of the switch S23 and one end of the switch S24.

[0074] Repeating a first phase and a second phase enables charging and discharging to be performed complementally on the capacitors C11 and C12.

[0075] Specifically, in the first phase, the switches S12, S13, S22, and S23 are turned ON. Accordingly, the one of the two electrodes of the capacitor C11 is connected to the node N3, the other of the two electrodes of the capacitor C11 and the one of the two electrodes of the capacitor C12 are connected to the node N2, and the other of the two electrodes of the capacitor C12 is connected to the node N1.

[0076] In contrast, in the second phase, the switches S11, S14, S21, and S24 are turned ON. Accordingly, the one of the two electrodes of the capacitor C12 is connected to the node N3, the other of the two electrodes of the capacitor C12 and the one of the two electrodes of the capacitor C11 are connected to the node N2, and the other of the two electrodes of the capacitor C11 is connected to the node N1.

[0077] Repeating the first phase and the second phase as described above enables, for example, when one of the capacitors C11 and C12 is charged through the node N1, the other of the capacitors C11 and C12 to be discharged to the capacitor C20. Charging and discharging may thus be performed complementally on the capacitors C11 and C12.

[0078] The capacitors C20 and C30 each function as a smoothing capacitor. Specifically, the capacitors C20 and C30 are used to hold and smooth the first to third output voltages (V3, V1, and V2) at the nodes N1 to N3.

[0079] The capacitor C20 is connected between the nodes N2 and N1. Specifically, one of the two electrodes of the capacitor C20 is connected to the node N2. In contrast, the other of the two electrodes of the capacitor C20 is connected to the node N1.

[0080] The capacitor C30 is connected between the nodes N3 and N2. Specifically, one of the two electrodes of the capacitor C30 is connected to the node N3. In contrast, the other of the two electrodes of the capacitor C30 is connected to the node N2.

[0081] The switch S11 is connected between the one of the two electrodes of the capacitor C11 and the node N2. Specifically, the one end of the switch S11 is connected to the one of the two electrodes of the capacitor C11. In contrast, the other end of the switch S11 is connected to the node N2.

[0082] The switch S12 is connected between the one of the two electrodes of the capacitor C11 and the node N3. Specifically, the one end of the switch S12 is connected to the one of the two electrodes of the capacitor C11. In contrast, the other end of the switch S12 is connected to the node N3.

[0083] The switch S21 is connected between the other of the two electrodes of the capacitor C11 and the node N1. Specifically, the one end of the switch S21 is connected to the other of the two

electrodes of the capacitor **C11**. In contrast, the other end of the switch **S21** is connected to the node **N1**.

[0084] The switch **S22** is connected between the other of the two electrodes of the capacitor **C11** and the node **N2**. Specifically, the one end of the switch **S22** is connected to the other of the two electrodes of the capacitor **C11**. In contrast, the other end of the switch **S22** is connected to the node **N2**.

[0085] The switch **S13** is connected between the one of the two electrodes of the capacitor **C12** and the node **N2**. Specifically, the one end of the switch **S13** is connected to the one of the two electrodes of the capacitor **C12**. In contrast, the other end of the switch **S13** is connected to the node **N2**. The other end of the switch **S13** is thus connected to the other end of the switch **S11** and the other end of the switch **S22**.

[0086] The switch **S14** is connected between the one of the two electrodes of the capacitor **C12** and the node **N3**. Specifically, the one end of the switch **S14** is connected to the one of the two electrodes of the capacitor **C12**. In contrast, the other end of the switch **S14** is connected to the node **N3**. The other end of the switch **S14** is thus connected to the other end of the switch **S12**.

[0087] The switch **S23** is connected between the other of the two electrodes of the capacitor **C12** and the node **N1**. Specifically, the one end of the switch **S23** is connected to the other of the two electrodes of the capacitor **C12**. In contrast, the other end of the switch **S23** is connected to the node **N1**. The other end of the switch **S23** is thus connected to the other end of the switch **S21**.

[0088] The switch **S24** is connected between the other of the two electrodes of the capacitor **C12** and the node **N2**. Specifically, the one end of the switch **S24** is connected to the other of the two electrodes of the capacitor **C12**. In contrast, the other end of the switch **S24** is connected to the node **N2**. The other end of the switch **S24** is thus connected to the other end of the switch **S11**, the other end of the switch **S22**, and the other end of the switch **S13**.

[0089] A first set of switches including the switches **S12**, **S13**, **S22**, and **S23** and a second set of switches including the switches **S11**, **S14**, **S21**, and **S24** are turned ON and OFF complementally based on a control signal from the digital control circuit **60**. Specifically, in the first phase, the switches in the first set are turned ON whereas the switches in the second set are turned OFF. Conversely, in the second phase, the switches in the first set are turned OFF whereas the switches in the second set are turned ON.

[0090] For example, in one of the first phase and the second phase, charging from the capacitor **C11** to the capacitors **C20** and **C30** is performed, and in the other of the first phase and the second phase, charging from the capacitor **C12** to the capacitors **C20** and **C30** is performed. The capacitors **C20** and **C30** are thus constantly charged by the capacitor **C11** or the capacitor **C12**. Accordingly, even when current flows rapidly from the nodes **N1** to **N3** to the supply modulator **30**, the nodes **N1** to **N3** are rapidly replenished with electric charges, and potential variations at the nodes **N1** to **N3** is prevented.

[0091] The operations as described above enable the switched-capacitor circuit **20** to maintain substantially equal voltages at both ends of each of the capacitors **C20** and **C30**. Specifically, the voltages **V1** to **V3** (voltages with respect to a ground potential) satisfying $(V3-V2):(V2-V1)=1:1$ are maintained at the four nodes labeled **V1** to **V3**. The voltages **V1** to **V3** correspond to a plurality of discrete voltages suppliable to the supply modulator **30** by the switched-capacitor circuit **20**.

[0092] The relation $(V3-V2):(V2-V1)$ is not limited to 1:1 and may be designed to be any ratio (such as 2:1, 3:1, 3:2, 1:2, or 2:3) according to various exemplary aspects.

[0093] The configuration of the switched-capacitor circuit **20** illustrated in FIG. 3 is illustrative and is not restrictive. For example, the switched-capacitor circuit **20** is configured to supply voltages of three discrete voltage levels but may be configured to supply voltages of any number of discrete voltage levels as long as the number is greater than or equal to four.

[1.2.2 Circuit Configuration of Supply Modulator **30**]

[0094] The circuit configuration of the supply modulator **30** will then be described with reference

to FIG. 3. The supply modulator **30** includes input terminals **131** to **133**, switches **S51** to **S53**, and an output terminal **134**.

[0095] The input terminals **131** to **133** are terminals for receiving the first to third output voltages of the switched-capacitor circuit **20**. The input terminals **131** to **133** are respectively connected to the output terminals **123** to **125** of the switched-capacitor circuit **20** outside the supply modulator **30** and are respectively connected to the switches **S51** to **S53** inside the supply modulator **30**.

[0096] The output terminal **134** is a terminal for selectively supplying at least one of a plurality of discrete voltages to the power amplifier **2**. The output terminal **134** is connected to the power amplifier **2** outside the supply modulator **30** and is connected to the switches **S51** to **S53** inside the supply modulator **30**.

[0097] The switch **S51** is connected between the input terminal **131** and the output terminal **134**. Specifically, the switch **S51** includes a terminal connected to the input terminal **131** and a terminal connected to the output terminal **134**. In this connection configuration, ON/OFF switching of the switch **S51** by a control signal from the digital control circuit **60** enables switching between connection and disconnection between the input terminal **131** and the output terminal **134**.

[0098] The switch **S52** is connected between the input terminal **132** and the output terminal **134**. Specifically, the switch **S52** includes a terminal connected to the input terminal **132** and a terminal connected to the output terminal **134**. In this connection configuration, ON/OFF switching of the switch **S52** by a control signal from the digital control circuit **60** enables switching between connection and disconnection between the input terminal **132** and the output terminal **134**.

[0099] The switch **S53** is connected between the input terminal **133** and the output terminal **134**. Specifically, the switch **S53** includes a terminal connected to the input terminal **133** and a terminal connected to the output terminal **134**. In this connection configuration, ON/OFF switching of the switch **S53** by a control signal from the digital control circuit **60** enables switching between connection and disconnection between the input terminal **133** and the output terminal **134**.

[0100] In this embodiment, these switches **S51** to **S53** are controlled to be exclusively turned ON. That is, only any one of the switches **S51** to **S53** is turned ON, and the others are turned OFF. The supply modulator **30** is thereby able to output one voltage selected from among the first to third output voltages of the switched-capacitor circuit **20** to the power amplifier **2**.

[0101] The configuration of the supply modulator **30** illustrated in FIG. 3 is illustrative and is not restrictive. In particular, the switches **S51** to **S53** may have any configuration and may be controlled in any manner as long as at least one of the three input terminals **131** to **133** are allowed be selectively connected to the output terminal **134**. For example, two of the switches **S51** to **S53** may be turned ON, and the other one of the switches **S51** to **S53** may be turned OFF.

[0102] In a case where four or more discrete voltages are supplied from the switched-capacitor circuit **20**, the supply modulator **30** may further include one or more switches in addition to the switches **S51** to **S53**.

[1.2.3 Circuit Configuration of Digital Control Circuit **60**]

[0103] The circuit configuration of the digital control circuit **60** will then be described with reference to FIG. 3. The digital control circuit **60** includes a first controller **61**, a second controller **62**, and control terminals **161** to **164**.

[0104] According to an exemplary aspect, the first controller **61** is configured to process source-synchronous digital control signals received from the RFIC **5** via the control terminals **161** and **162** and thereby generate a control signal for controlling the switched-capacitor circuit **20**. Turning ON/OFF of the switches **S11** to **S14** and **S21** to **S24** included in the switched-capacitor circuit **20** is controlled by the control signal from the first controller **61**.

[0105] The digital control signals processed by the first controller **61** are not limited to source-synchronous digital control signals. For example, the first controller **61** may process clock-embedded digital control signals. The first controller **61** may generate a control signal for controlling the supply modulator **30**.

[0106] Although one set of a clock signal and a data signal is used in this exemplary embodiment, the exemplary aspects of the present disclosure are so not limited thereto. For example, a plurality of sets of a clock signal and a data signal may be used as digital control signals.

[0107] The second controller **62** processes digital control logic/line (DCL) signals (DCL1 and DCL2) received from the RFIC **5** via the control terminals **163** and **164** and thereby generates a control signal for controlling the supply modulator **30**. The DCL signals (DCL1 and DCL2) are generated by the RFIC **5** based on an envelope signal of a radio frequency signal. Turning ON/OFF of the switches S51 to S53 included in the supply modulator **30** is controlled by the control signal from the second controller **62**.

[0108] The DCL signals (DCL1 and DCL2) are each a 1-bit signal. The voltages V1 to V3 are each represented by a combination of two 1-bit signals. For example, V1, V2, and V3 are represented by "00", "01", and "10", respectively. A gray code may be used to express a voltage level.

[0109] In this embodiment, two digital control logic/line signals are used to control the supply modulator **30**, but the number of digital control logic/line signals is not limited thereto. For example, any number of digital control logic/line signals may be used as long as the number is one, three, or more, in accordance with the number of voltage levels selectable by the respective switches of the supply modulator **30**. The digital control signal used to control the supply modulator **30** is not limited to a digital control logic/line signal.

[1.3 Circuit Configuration of Pre-Regulator Circuit **10**]

[0110] The circuit configuration of the pre-regulator circuit **10** will then be described with reference to FIG. 3. The pre-regulator circuit **10** includes an input terminal **110**, output terminals **111** and **112**, switches S71 to S75, a power inductor L71, and capacitors C71 and C72.

[0111] The input terminal **110** is a terminal for receiving a power supply voltage from the direct current power source **50**. The input terminal **110** is connected to the direct current power source **50** outside the pre-regulator circuit **10** and is connected to the switch S71 inside the pre-regulator circuit **10**.

[0112] The output terminal **111** is an example of a first output terminal and is a terminal for supplying a first input voltage (V3) to the switched-capacitor circuit **20**. The output terminal **111** is connected to the input terminal **121** of the switched-capacitor circuit **20** outside the pre-regulator circuit **10** and is connected to the switch S73 inside the pre-regulator circuit **10**.

[0113] The output terminal **112** is an example of a second output terminal and is a terminal for supplying a second input voltage (V1) to the switched-capacitor circuit **20**. The output terminal **112** is connected to the input terminal **122** of the switched-capacitor circuit **20** outside the pre-regulator circuit **10** and is connected to the switch S74 inside the pre-regulator circuit **10**.

[0114] The power inductor L71 is an inductor used to raise and lower a direct current voltage. One end of the power inductor L71 is connected to the switches S71 and S72, and the other end of the power inductor L71 is connected to the switches S73 to S75.

[0115] The switch S71 is connected between the input terminal **110** and the one end of the power inductor L71. Specifically, the switch S71 includes a terminal connected to the input terminal **110** and a terminal connected to the one end of the power inductor L71. In this connection configuration, ON/OFF switching of the switch S71 enables switching between connection and disconnection between the input terminal **110** and the one end of the power inductor L71.

[0116] The switch S72 is connected between the one end of the power inductor L71 and ground. Specifically, the switch S72 includes a terminal connected to the one end of the power inductor L71 and a terminal connected to ground. In this connection configuration, ON/OFF switching of the switch S72 enables switching between connection and disconnection between the one end of the power inductor L71 and ground.

[0117] The switch S73 is an example of a first switch and is connected between the other end of the power inductor L71 and the output terminal **111**. Specifically, the switch S73 includes a terminal connected to the other end of the power inductor L71 and a terminal connected to the output

terminal **111**. In this connection configuration, ON/OFF switching of the switch **S73** enables switching between connection and disconnection between the other end of the power inductor **L71** and the output terminal **111**.

[0118] The switch **S74** is an example of a second switch and is connected between the other end of the power inductor **L71** and the output terminal **112**. Specifically, the switch **S74** includes a terminal connected to the other end of the power inductor **L71** and a terminal connected to the output terminal **112**. In this connection configuration, ON/OFF switching of the switch **S74** enables switching between connection and disconnection between the other end of the power inductor **L71** and the output terminal **112**.

[0119] The switches **S73** and **S74** each have a withstand voltage lower than the withstand voltage of each of the switches **S11** to **S14** and **S21** to **S24** included in the switched-capacitor circuit **20**. Conversely, the switches **S11** to **S14** and **S21** to **S24** included in the switched-capacitor circuit **20** each have a withstand voltage higher than the withstand voltage of each of the switches **S73** and **S74** included in the pre-regulator circuit **10**.

[0120] It is possible to compare withstand voltages of two switches with each other based on the number of transistors connected in series thereto. Specifically, the withstand voltage of a switch connected in series to a larger number of transistors is higher than the withstand voltage of a switch connected in series to a smaller number of transistors.

[0121] The switch **S75** is an example of a third switch and is connected between the other end of the power inductor **L71** and ground. Specifically, the switch **S75** includes a terminal connected to the other end of the power inductor **L71** and a terminal connected to ground. In this connection configuration, ON/OFF switching of the switch **S75** enables switching between connection and disconnection between the other end of the power inductor **L71** and ground.

[0122] The capacitor **C71** is an example of a first capacitor and is connected between ground and a path between the switch **S73** and the output terminal **111**. Specifically, one of the two electrodes of the capacitor **C71** is connected to the switch **S73** and the output terminal **111**, and the other of the two electrodes of the capacitor **C71** is connected to ground.

[0123] The capacitor **C72** is an example of a second capacitor and is connected between ground and a path between the switch **S74** and the output terminal **112**. Specifically, one of the two electrodes of the capacitor **C72** is connected to the switch **S74** and the output terminal **112**, and the other of the two electrodes of the capacitor **C72** is connected to ground.

[0124] According to an exemplary aspect, the pre-regulator circuit **10** having the above-described configuration is configured to convert a power supply voltage into two variable voltages independent of each other (the first input voltage and the second input voltage of the switched-capacitor circuit **20**).

[0125] The configuration of the pre-regulator circuit **10** illustrated in FIG. **3** is illustrative and is not restrictive. For example, one or some of the switches **S71** to **S75** may be replaced with a diode. Part or the entirety of the pre-regulator circuit **10** may be included in the tracker circuit **1**.

[1.4 Tracking Method]

[0126] A tracking method that is a method for supplying a plurality of discrete voltages by the tracker circuit **1** having the above-described configuration will then be described with reference to FIG. **4**, FIG. **5A** to FIG. **5C**, and FIG. **6A** to FIG. **6C**. FIG. **4** is a flowchart illustrating the tracking method according to this exemplary embodiment. FIG. **5A** is a graph illustrating a radio frequency signal **RF1** modulated by QPSK and a plurality of discrete voltages. FIG. **5B** is a graph illustrating a relationship between the output power of the power amplifier that amplifies the radio frequency signal **RF1** modulated by QPSK and an occurrence frequency. FIG. **5C** is a graph illustrating a relationship between the output power of the power amplifier **2** that amplifies the radio frequency signal **RF1** modulated by QPSK and power efficiency. FIG. **6A** is a graph illustrating a radio frequency signal **RF2** modulated by 256QAM and a plurality of discrete voltages. FIG. **6B** is a graph illustrating a relationship between the output power of the power amplifier **2** that amplifies

the radio frequency signal RF2 modulated by 256QAM and an occurrence frequency. FIG. 6C is a graph illustrating a relationship between the output power of the power amplifier 2 that amplifies the radio frequency signal RF2 modulated by 256QAM and power efficiency.

<Step S10>

[0127] The switched-capacitor circuit 20 receives a first input voltage and a second input voltage from the pre-regulator circuit 10. For example, when a radio frequency signal RF1 modulated by discrete Fourier transform spreading-orthogonal frequency division multiplexing quadrature phase shift keying (DFTs-OFDM QPSK, hereinafter simply referred to as QPSK) is amplified by the power amplifier 2, the switched-capacitor circuit 20 receives a first input voltage ($V_{\max1}$) and a second input voltage ($V_{\min1}$) from the pre-regulator circuit 10. QPSK is an example of a modulation scheme having a lower bit rate.

<Step S20>

[0128] The switched-capacitor circuit 20 generates a plurality of first discrete voltages based on a first difference between the first input voltage and the second input voltage. Specifically, the switched-capacitor circuit 20 generates a first output voltage based on the first input voltage, a second output voltage based on the second input voltage, and a third output voltage based on the voltage difference between the first input voltage and the second input voltage. For example, in a case where the radio frequency signal RF1 modulated by QPSK is amplified by the power amplifier 2, the switched-capacitor circuit 20 generates, as a plurality of first discrete voltages, a first output voltage ($V_{\max1}$) equal to the first input voltage, a second output voltage ($V_{\min1}$) equal to the second input voltage, and a third output voltage ($V_{\text{mid}1} = V_{\max1} - V_{d1} = V_{\min1} + V_{d1}$) that is lower than the first input voltage by a half (V_{d1}) of the first difference and higher than the second input voltage by a half (V_{d1}) of the first difference, based on the first difference ($2V_{d1}$) between the first input voltage ($V_{\max1}$) and the second input voltage ($V_{\min1}$).

<Step S30>

[0129] The supply modulator 30 selectively outputs at least one of the plurality of first discrete voltages to the power amplifier 2. For example, in a case where the radio frequency signal RF1 modulated by QPSK is amplified by the power amplifier 2, the supply modulator 30 selectively outputs the first output voltage ($V_{\max1}$), the second output voltage ($V_{\min1}$), and the third output voltage ($V_{\text{mid}1}$) to the power amplifier 2, based on an envelope signal. As illustrated in FIG. 5A, a voltage based on the envelope of the radio frequency signal RF1 is thereby supplied to the power amplifier 2.

[0130] Steps S10 to S30 as described above enables the first output voltage ($V_{\max1}$) corresponding to the maximum output power of the radio frequency signal RF1 to be output and the second output voltage ($V_{\min1}$) corresponding to output power $P_{\text{out}1}$ with the highest occurrence frequency of the radio frequency signal RF1 to be output, as illustrated in FIG. 5B and FIG. 5C. This enables the maximum output power of the radio frequency signal RF1 to be supported and the power efficiency for the output power with the highest occurrence frequency of the radio frequency signal RF1 to be improved. Furthermore, the third output voltage ($V_{\text{mid}1}$) as a middle between the first output voltage ($V_{\max1}$) and the second output voltage ($V_{\min1}$) may also be output, and thus the power efficiency for the output power between the maximum output power and an average output voltage may also be improved.

<Step S40>

[0131] The switched-capacitor circuit 20 receives a third input voltage and a fourth input voltage from the pre-regulator circuit 10. A second difference between the third input voltage and the fourth input voltage is different from the first difference between the first input voltage and the second input voltage. For example, in a case where a radio frequency signal RF2 modulated by cyclic prefix-orthogonal frequency division multiplexing 256 quadrature amplitude modulation (CP-OFDM 256QAM, hereinafter simply referred to as 256QAM) is amplified by the power amplifier 2, the switched-capacitor circuit 20 receives a third input voltage ($V_{\max2}$) and a fourth input

voltage (V_{min2}) from the pre-regulator circuit **10**. At this time, the bit rate of 256QAM is higher than the bit rate of QPSK, and thus the second difference ($2V_{d2}$) is larger than the first difference ($2V_{d1}$).

[0132] It should be appreciated that 256QAM is an example of a modulation scheme having a higher bit rate. In general, QAM (such as 256QAM, 64QAM, or 16QAM) has a higher bit rate and a higher peak to average power ratio (PAPR) than PSK (such as QPSK or binary phase-shift keying (BPSK)).

<Step S50>

[0133] The switched-capacitor circuit **20** generates a plurality of second discrete voltages based on the second difference between the third input voltage and the fourth input voltage. The second difference between the third input voltage and the fourth input voltage is different from the first difference between the first input voltage and the second input voltage, and thus the plurality of second discrete voltages is different from the plurality of first discrete voltages. For example, in a case where the radio frequency signal RF2 modulated by 256QAM is amplified by the power amplifier **2**, the switched-capacitor circuit **20** generates, as the plurality of second discrete voltages, a fourth output voltage (V_{max2}) equal to the third input voltage, a fifth output voltage (V_{min2}) equal to the fourth input voltage, and a sixth output voltage ($V_{mid2}=V_{max2}-V_{d2}=V_{min2}+V_{d2}$) that is lower than the third input voltage by a half (V_{d2}) of the second difference and higher than the fourth input voltage by a half (V_{d2}) of the second difference, based on the second difference ($2V_{d2}$) between the third input voltage (V_{max2}) and the fourth input voltage (V_{min2}).

<Step S60>

[0134] The supply modulator **30** selectively outputs at least one of the plurality of second discrete voltages to the power amplifier **2**. For example, in a case where the radio frequency signal RF2 modulated by 256QAM is amplified by the power amplifier **2**, the supply modulator **30** selectively outputs the fourth output voltage (V_{max2}), the fifth output voltage (V_{min2}), and the sixth output voltage (V_{mid2}) to the power amplifier **2**, based on an envelope signal. As illustrated in FIG. 6A, a voltage based on the envelope of the radio frequency signal RF2 is thereby supplied to the power amplifier **2**.

[0135] Steps S40 to S60 as described above enables the fourth output voltage (V_{max2}) corresponding to the maximum output power of the radio frequency signal RF2 to be output and the fifth output voltage (V_{min2}) corresponding to output power P_{out2} with the highest occurrence frequency of the radio frequency signal RF2 to be output, as illustrated in FIG. 6B and FIG. 6C. This enables the maximum output power of the radio frequency signal RF2 to be supported and the power efficiency for the output power with the highest occurrence frequency of the radio frequency signal RF2 to be improved. Furthermore, the sixth output voltage (V_{mid2}) as a middle between the fourth output voltage (V_{max2}) and the fifth output voltage (V_{min2}) may also be output, and thus the power efficiency for the output power between the maximum output power and an average output voltage may also be improved.

[0136] As described above, different input voltages are used for radio frequency signals that vary in characteristics (in particular, PAPR), and thereby varying the modulation scheme enables generation of an output voltage corresponding to the output power with the highest occurrence frequency (the second output voltage and the fifth output voltage) and an output voltage corresponding to the maximum output power (the first output voltage and the fourth output voltage). As the result, even in exemplary aspects when the number of the plurality of discrete voltages generated by the switched-capacitor circuit **20** is reduced, a decrease in the power efficiency of the power amplifier **2** is prevented or minimized, and the configuration of the switched-capacitor circuit **20** is simplified.

[1.5 Technical Effects and the Like]

[0137] As described above, the tracker circuit **1** according to this embodiment includes the switched-capacitor circuit **20** configured to generate a plurality of discrete voltages and the supply

modulator **30** configured to selectively output at least one of the plurality of discrete voltages to the power amplifier **2**. The switched-capacitor circuit **20** is configured to generate the plurality of discrete voltages based on a difference between a first input voltage and a second input voltage, the second input voltage being variable and lower than the first input voltage.

[0138] The switched-capacitor circuit thereby generates the plurality of discrete voltages based on the voltage difference between the first input voltage and the second input voltage. Accordingly, the plurality of discrete voltages may be set more flexibly than in a case where the plurality of discrete voltages are generated based on one input voltage, and the power efficiency of the power amplifier **2** may be improved. Furthermore, even in exemplary aspects when the number of the plurality of discrete voltages generated by the switched-capacitor circuit **20** (i.e., the number of levels) is reduced, a decrease in the power efficiency of the power amplifier **2** may be prevented by changing the second input voltage. Accordingly, the configuration of the switched-capacitor circuit **20** is simplified, and the power efficiency of the tracker circuit **1** is improved.

[0139] For example, in the tracker circuit **1** according to this embodiment, the first input voltage may be variable.

[0140] Accordingly, since both the first input voltage and the second input voltage are variable, the plurality of discrete voltages may be set more flexibly. Accordingly, the power efficiency of the power amplifier **2** is further improved, and the configuration of the switched-capacitor circuit **20** is further simplified.

[0141] For example, in the tracker circuit **1** according to this embodiment, the plurality of discrete voltages may include a first output voltage based on the first input voltage, a second output voltage based on the second input voltage, and a third output voltage based on the voltage difference between the first input voltage and the second input voltage.

[0142] Accordingly, it suffices that only the third output voltage among the first output voltage, the second output voltage, and the third output voltage be generated based on the voltage difference between the first input voltage and the second input voltage. Power consumption for generating the first output voltage and the second output voltage may be reduced, and the plurality of discrete voltages may be generated more stably.

[0143] For example, in the tracker circuit **1** according to this embodiment, the third output voltage may be lower than the first output voltage.

[0144] Accordingly, the power efficiency of the switched-capacitor circuit **20** is increased, and the plurality of discrete voltages are generated more stably.

[0145] For example, in the tracker circuit **1** according to this embodiment, the switched-capacitor circuit **20** may be configured to generate the third output voltage to cause each of a difference between the first output voltage and the third output voltage and a difference between the second output voltage and the third output voltage to increase in response to an increase in the voltage difference between the first input voltage and the second input voltage.

[0146] The step widths of the plurality of discrete voltages may thereby be controlled by the voltage difference between the first input voltage and the second input voltage. Accordingly, a plurality of discrete voltages suitable for the characteristics (such as PAPR) of a radio frequency signal may be generated, and power efficiency is improved.

[0147] For example, in the tracker circuit **1** according to this embodiment, the voltage difference between the first input voltage and the second input voltage may be switched in accordance with a modulation scheme for a radio frequency signal amplified by the power amplifier **2**.

[0148] In a case where the characteristics (such as PAPR) of the radio frequency signal change according to a modulation scheme, this enables the step widths of the plurality of discrete voltages to be changed and power efficiency to be improved.

[0149] For example, in the tracker circuit **1** according to this embodiment, the voltage difference between the first input voltage and the second input voltage may increase as a bit rate of the modulation scheme for the radio frequency signal increases.

[0150] Accordingly, the step widths of the plurality of discrete voltages may be widened as the bit rate of the modulation scheme increases, and it is possible to cope with the PAPR which increases with an increase in the bit rate and to improve power efficiency.

[0151] For example, in the tracker circuit **1** according to this embodiment, the voltage difference between the first input voltage and the second input voltage may be switched at the time of switching of a frame of the radio frequency signal amplified by the power amplifier **2**.

[0152] Accordingly, it is possible to prevent radio frequency signal deterioration due to instability of the plurality of discrete voltages at the time of switching of the voltage difference between the first input voltage and the second input voltage.

[0153] For example, the tracker circuit **1** according to this embodiment may be connected to the pre-regulator circuit **10** configured to convert a power supply voltage into the first input voltage and the second input voltage.

[0154] Accordingly, since the tracker circuit **1** is connected to the pre-regulator circuit **10** having one input and two outputs, the number of pre-regulator circuits **10** may be reduced.

[0155] The communication device **7** according to this embodiment includes the tracker circuit **1**, the pre-regulator circuit **10** configured to convert a power supply voltage into the first input voltage and the second input voltage, the RFIC **5** configured to process a radio frequency signal, and the radio frequency circuit **4** including the power amplifier **2** and configured to transmit the radio frequency signal between the RFIC **5** and the antenna **6**.

[0156] It is noted that technical effects similar to those of the tracker circuit **1** may thereby be achieved. As the result, the power consumption of the communication device **7** is reduced, and the battery life of the communication device **7** can be extended.

[0157] A tracking method according to this exemplary embodiment includes receiving a first input voltage and a second input voltage lower than the first input voltage (**S10**); generating a plurality of first discrete voltages based on a first difference between the first input voltage and the second input voltage (**S20**); selectively outputting at least one of the plurality of first discrete voltages to the power amplifier **2** (**S30**); receiving a third input voltage and a fourth input voltage that is lower than the third input voltage (**S40**), the fourth input voltage being different from the second input voltage; generating a plurality of second discrete voltages based on a second difference between the third input voltage and the fourth input voltage (**S50**); and selectively outputting at least one of the plurality of second discrete voltages to the power amplifier **2** (**S60**).

[0158] This causes the plurality of first discrete voltages to be generated based on the first difference between the first input voltage and the second input voltage and the plurality of second discrete voltages to be generated based on the second difference between the third input voltage and the fourth input voltage. Accordingly, changing two input voltages enables the plurality of discrete voltages to be set more flexibly than in the case where the plurality of discrete voltages are generated based on one input voltage and enables the power efficiency of the power amplifier **2** to be improved. Furthermore, even in an exemplary aspect when the number of the plurality of discrete voltages is reduced, a decrease in the power efficiency of the power amplifier **2** may be prevented by changing the second input voltage. Accordingly, the configuration of the tracker circuit **1** is simplified, and the power efficiency of the tracker circuit **1** is improved.

Modification of Exemplary Embodiment 1

[0159] A modification of Embodiment 1 will then be described. As described for Embodiment 1 above, the tracker circuit **1** may include part or the entirety of the pre-regulator circuit **10**. Hence, a tracker circuit **1** according to this modification includes the entire pre-regulator circuit **10**, as illustrated in FIG. 7. The tracker circuit **1** may include only part of the pre-regulator circuit **10**, for example, only the capacitors **C71** and **C72**.

[0160] As described above, the tracker circuit **1** according to this modification includes the switched-capacitor circuit **20** including the input terminal **121** and the input terminal **122**, the supply modulator **30**, the capacitor **C71**, and the capacitor **C71**. The switched-capacitor circuit **20** is

configured to generate the plurality of discrete voltages. The input terminal **121** receives the first input voltage, and the input terminal **122** receives the second input voltage that is variable and that is lower than the first input voltage. The supply modulator **30** is configured to selectively output at least one of the plurality of discrete voltages to the power amplifier **2**. The capacitor **C71** is connected between the input terminal **121** and ground, and the capacitor **C72** is connected between the input terminal **122** and ground.

[0161] The first input voltage and the second input voltage may thereby be received via the input terminals **121** and **122** which are connected in shunt to the two different capacitors **C71** and **C72**, and thus the switched-capacitor circuit **20** may receive the first input voltage and the second input voltage independent of each other. Accordingly, the plurality of discrete voltages may be set more flexibly than in the case where the plurality of discrete voltages are generated based on one input voltage, and the power efficiency of the power amplifier **2** may be improved. Furthermore, even in an exemplary aspect when the number of plurality of discrete voltages generated by the switched-capacitor circuit **20** (i.e., the number of levels) is reduced, a decrease in the power efficiency of the power amplifier **2** is prevented or minimized. Accordingly, the configuration of the switched-capacitor circuit **20** is simplified, and the power efficiency of the tracker circuit **1** is also improved.

[0162] For example, the tracker circuit **1** according to this modification may further include the pre-regulator circuit **10**. The pre-regulator circuit **10** may include the power inductor **L71**, the output terminal **111** connected to the input terminal **121** of the switched-capacitor circuit **20**, the output terminal **112** connected to the input terminal **122** of the switched-capacitor circuit **20**, the switch **S73** connected between an output end of the power inductor **L71** and the output terminal **111**, the switch **S74** connected between the output end of the power inductor **L71** and the output terminal **112**, the switch **S75** connected between the output end of the power inductor **L71** and ground, and the capacitor **C71** and the capacitor **C72**. The capacitor **C71** may be connected between ground and a path between the switch **S73** and the output terminal **111**. The capacitor **C72** may be connected between ground and a path between the switch **S74** and the output terminal **112**.

[0163] Since the tracker circuit **1** thereby includes the one-input two-output pre-regulator circuit **10**, it is not necessary to provide a pre-regulator circuit **10** separately, and the tracker circuit **1** may be easily introduced into the communication device **7**.

[0164] For example, in the tracker circuit **1** according to this modification, the withstand voltage of a switch included in the switched-capacitor circuit **20** may be higher than the withstand voltage of each of the switches **S73** and **S74** included in the pre-regulator circuit **10**.

[0165] This configuration enables the withstand voltage of the switch included in the switched-capacitor circuit **20** to be made relatively high, breakdown of the switched-capacitor circuit **20** to be prevented in a case where the voltage difference between the first input voltage and the second input voltage is large, and the plurality of discrete voltages to be set more flexibly.

Exemplary Embodiment 2

[0166] Embodiment 2 will then be described. This embodiment is different from Embodiment 1 described above mainly in that a switched-capacitor circuit receives two input voltages from two pre-regulator circuits, respectively. Hereinafter, this embodiment will be described with reference to the drawings, with a focus on the difference from Embodiment 1 described above.

[2.1 Circuit Configuration of Communication Device 7A]

[0167] A communication device **7A** according to this embodiment will be described with reference to FIG. **8**. FIG. **8** is a circuit configuration diagram of the communication device **7A** according to this embodiment.

[0168] FIG. **8** illustrates an exemplary circuit configuration, and the communication device **7A** may be implemented by using any of a wide variety of circuit implementation and circuit techniques. Accordingly, the description of the communication device **7A** provided below should not be construed in a limiting manner.

[0169] The communication device **7A** according to this embodiment includes the tracker circuit **1**,

the radio frequency circuit **4**, the RFIC **5**, the antenna **6**, pre-regulator circuits **10A** and **10B**, and the direct current power source **50**.

[0170] According to an exemplary aspect, the pre-regulator circuit **10A** is an example of a first converter circuit and is a one-input one-output buck-boost converter. The pre-regulator circuit **10A** is configured to convert a power supply voltage supplied from the direct current power source **50** into one variable voltage (the first input voltage of the switched-capacitor circuit **20**). The pre-regulator circuit **10A** is configured to change the first input voltage based on, for example, a control signal from the RFIC **5**. The circuit configuration of the pre-regulator circuit **10A** will be described later by using FIG. **9**.

[0171] The pre-regulator circuit **10B** is an example of a second converter circuit and is a one-input one-output buck-boost converter. The pre-regulator circuit **10B** is configured to convert a power supply voltage supplied from the direct current power source **50** into one variable voltage (the second input voltage of the switched-capacitor circuit **20**). The pre-regulator circuit **10B** is configured to change the second input voltage based on, for example, a control signal from the RFIC **5**. The circuit configuration of the pre-regulator circuit **10B** will be described later by using FIG. **9**.

[0172] According to an exemplary aspect, the pre-regulator circuits **10A** and **10B** are configured to switch the voltage difference between the first input voltage and the second input voltage in accordance with, for example, a modulation scheme for a radio frequency signal amplified by the power amplifier **2**. More specifically, the pre-regulator circuits **10A** and **10B** can be configured to increase the voltage difference between the first input voltage and the second input voltage as the bit rate of the modulation scheme increases. Such switching of the voltage difference between the first input voltage and the second input voltage may be performed at the time of switching of a frame of the radio frequency signal.

[2.2 Circuit Configuration of Pre-Regulator Circuit **10A**]

[0173] The circuit configuration of the pre-regulator circuit **10A** will be described with reference to FIG. **9**. FIG. **9** is a circuit configuration diagram of the tracker circuit **1** and the pre-regulator circuits **10A** and **10B** according to this embodiment.

[0174] FIG. **9** illustrates an exemplary circuit configuration, and the tracker circuit **1** and the pre-regulator circuits **10A** and **10B** may be implemented by using any of a wide variety of circuit implementation and circuit techniques. Accordingly, the description of the individual circuits provided below should not be construed in a limiting manner.

[0175] The pre-regulator circuit **10A** includes an input terminal **110A**, an output terminal **111A**, switches **S71A** to **S73A** and **S75A**, a power inductor **L71A**, and a capacitor **C71A**.

[0176] The input terminal **110A** is a terminal for receiving a power supply voltage from the direct current power source **50**. The input terminal **110A** is connected to the direct current power source **50** outside the pre-regulator circuit **10A** and is connected to the switch **S71A** inside the pre-regulator circuit **10A**.

[0177] The output terminal **111A** is a terminal for supplying a first input voltage (**V3**) to the switched-capacitor circuit **20**. The output terminal **111A** is connected to the input terminal **121** of the switched-capacitor circuit **20** outside the pre-regulator circuit **10A** and is connected to the switch **S73A** inside the pre-regulator circuit **10A**.

[0178] The power inductor **L71A** is an inductor used to raise and lower a direct current voltage. One end of the power inductor **L71A** is connected to the switches **S71A** and **S72A**, and the other end of the power inductor **L71A** is connected to the switches **S73A** and **S75A**.

[0179] The switch **S71A** is connected between the input terminal **110A** and the one end of the power inductor **L71A**. Specifically, the switch **S71A** includes a terminal connected to the input terminal **110A** and a terminal connected to the one end of the power inductor **L71A**. In this connection configuration, ON/OFF switching of the switch **S71A** enables switching between connection and disconnection between the input terminal **110A** and the one end of the power

inductor **L71A**.

[0180] The switch **S72A** is connected between the one end of the power inductor **L71A** and ground. Specifically, the switch **S72A** includes a terminal connected to the one end of the power inductor **L71A** and a terminal connected to ground. In this connection configuration, ON/OFF switching of the switch **S72A** enables switching between connection and disconnection between the one end of the power inductor **L71A** and ground.

[0181] The switch **S73A** is connected between the other end of the power inductor **L71A** and the output terminal **111A**. Specifically, the switch **S73A** includes a terminal connected to the other end of the power inductor **L71A** and a terminal connected to the output terminal **111A**. In this connection configuration, ON/OFF switching of the switch **S73A** enables switching between connection and disconnection between the other end of the power inductor **L71A** and the output terminal **111A**.

[0182] The switch **S75A** is connected between the other end of the power inductor **L71A** and ground. Specifically, the switch **S75A** includes a terminal connected to the other end of the power inductor **L71A** and a terminal connected to ground. In this connection configuration, ON/OFF switching of the switch **S75A** enables switching between connection and disconnection between the other end of the power inductor **L71A** and ground.

[0183] The capacitor **C71A** is connected between ground and a path between the switch **S73A** and the output terminal **111A**. Specifically, one of the two electrodes of the capacitor **C71A** is connected to the switch **S73A** and the output terminal **111A**, and the other of the two electrodes of the capacitor **C71A** is connected to ground.

[0184] According to an exemplary aspect, the pre-regulator circuit **10A** having the above-described configuration is configured to convert a power supply voltage into a variable voltage **V3** (e.g., the first input voltage of the switched-capacitor circuit **20**).

[0185] The configuration of the pre-regulator circuit **10A** illustrated in FIG. **9** is illustrative and is not restrictive. For example, one or some of the switches **S71A** to **S73A** and **S75A** may be replaced with a diode. Part or the entirety of the pre-regulator circuit **10A** may be included in the tracker circuit **1**, like the pre-regulator circuit **10** according to the modification of Embodiment 1.

[2.3 Circuit Configuration of Pre-Regulator Circuit **10B**]

[0186] The circuit configuration of the pre-regulator circuit **10B** will be described with reference to FIG. **9**. The pre-regulator circuit **10B** includes an input terminal **110B**, an output terminal **112B**, switches **S71B**, **S72B**, **S74B**, and **S75B**, a power inductor **L71B**, and a capacitor **C72B**.

[0187] The input terminal **110B** is a terminal for receiving a power supply voltage from the direct current power source **50**. The input terminal **110B** is connected to the direct current power source **50** outside the pre-regulator circuit **10B** and is connected to the switch **S71B** inside the pre-regulator circuit **10B**.

[0188] The output terminal **112B** is a terminal for supplying a second input voltage (**V1**) to the switched-capacitor circuit **20**. The output terminal **112B** is connected to the input terminal **122** of the switched-capacitor circuit **20** outside the pre-regulator circuit **10B** and is connected to the switch **S74B** inside the pre-regulator circuit **10B**.

[0189] The power inductor **L71B** is an inductor used to raise and lower a direct current voltage. One end of the power inductor **L71B** is connected to the switches **S71B** and **S72B**, and the other end of the power inductor **L71B** is connected to the switches **S74B** and **S75B**.

[0190] The switch **S71B** is connected between the input terminal **110B** and the one end of the power inductor **L71B**. Specifically, the switch **S71B** includes a terminal connected to the input terminal **110B** and a terminal connected to the one end of the power inductor **L71B**. In this connection configuration, ON/OFF switching of the switch **S71B** enables switching between connection and disconnection between the input terminal **110B** and the one end of the power inductor **L71B**.

[0191] The switch **S72B** is connected between the one end of the power inductor **L71B** and ground.

Specifically, the switch **S72B** includes a terminal connected to the one end of the power inductor **L71B** and a terminal connected to ground. In this connection configuration, ON/OFF switching of the switch **S72B** enables switching between connection and disconnection between the one end of the power inductor **L71B** and ground.

[0192] The switch **S74B** is connected between the other end of the power inductor **L71B** and the output terminal **112B**. Specifically, the switch **S74B** includes a terminal connected to the other end of the power inductor **L71B** and a terminal connected to the output terminal **112B**. In this connection configuration, ON/OFF switching of the switch **S74B** enables switching between connection and disconnection between the other end of the power inductor **L71B** and the output terminal **112B**.

[0193] The switch **S75B** is connected between the other end of the power inductor **L71B** and ground. Specifically, the switch **S75B** includes a terminal connected to the other end of the power inductor **L71B** and a terminal connected to ground. In this connection configuration, ON/OFF switching of the switch **S75B** enables switching between connection and disconnection between the other end of the power inductor **L71B** and ground.

[0194] The capacitor **C72B** is connected between ground and a path between the switch **S74B** and the output terminal **112B**. Specifically, one of the two electrodes of the capacitor **C72B** is connected to the switch **S74B** and the output terminal **112B**, and the other of the two electrodes of the capacitor **C72B** is connected to ground.

[0195] According to an exemplary aspect, the pre-regulator circuit **10B** having the above-described configuration is configured to convert a power supply voltage into a variable voltage **V1** (the second input voltage of the switched-capacitor circuit **20**).

[0196] The configuration of the pre-regulator circuit **10B** illustrated in FIG. **9** is illustrative and is not restrictive. For example, one or some of the switches **S71B**, **S72B**, **S74B**, and **S75B** may be replaced with a diode. Part or the entirety of the pre-regulator circuit **10B** may be included in the tracker circuit **1**, like the pre-regulator circuit **10** according to the modification of Embodiment 1.

[2.4 Technical Effects and the Like]

[0197] As described above, the tracker circuit **1** according to this embodiment is connected to the pre-regulator circuit **10A** and the pre-regulator circuit **10B**, the pre-regulator circuit **10A** being configured to convert a power supply voltage into the first input voltage, the pre-regulator circuit **10B** being configured to convert the power supply voltage into the second input voltage.

[0198] Accordingly, one-input one-output buck-boost converters, which are often used in communication devices in the related art, may be used, and the tracker circuit **1** may be easily introduced into the communication device **7A**.

Exemplary Embodiment 3

[0199] Embodiment 3 will then be described. This embodiment is different from Embodiments 1 and 2 described above mainly in that a voltage supplied from a pre-regulator circuit is lowered in a tracker circuit to thereby generate two input voltages of a switched-capacitor circuit. Hereinafter, this embodiment will be described with reference to the drawings, with a focus on the difference from Embodiments 1 and 2 described above.

[3.1 Circuit Configuration of Communication Device 7B]

[0200] A communication device **7B** according to this embodiment will be described with reference to FIG. **10**. FIG. **10** is a circuit configuration diagram of the communication device **7B** according to this embodiment.

[0201] FIG. **10** illustrates an exemplary circuit configuration, and the communication device **7B** may be implemented by using any of a wide variety of circuit implementation and circuit techniques. Accordingly, the description of the communication device **7B** provided below should not be construed in a limiting manner.

[0202] The communication device **7B** according to this embodiment includes a tracker circuit **1B**, the radio frequency circuit **4**, the RFIC **5**, the antenna **6**, the pre-regulator circuit **10A**, and the

direct current power source **50**.

[0203] According to an exemplary aspect, the tracker circuit **1B** is configured to supply a plurality of discrete voltages to the power amplifier **2** in the digital ET mode. As illustrated in FIG. **10**, the tracker circuit **1B** includes a pre-regulator circuit **10C**, the switched-capacitor circuit **20**, the supply modulator **30**, and a digital control circuit **60B**.

[0204] The pre-regulator circuit **10C** is an example of a converter circuit and is a buck converter. The pre-regulator circuit **10C** is configured to lower a first input voltage supplied from the pre-regulator circuit **10A** to a variable second input voltage. The pre-regulator circuit **10C** is configured to change the second input voltage based on a control signal from the digital control circuit **60**. The circuit configuration of the pre-regulator circuit **10C** will be described later by using FIG. **11**.

[0205] According to an exemplary aspect, the pre-regulator circuits **10A** and **10C** are configured to switch the voltage difference between the first input voltage and the second input voltage in accordance with, for example, a modulation scheme for a radio frequency signal amplified by the power amplifier **2**. More specifically, the pre-regulator circuits **10A** and **10C** can be configured to increase the voltage difference between the first input voltage and the second input voltage as the bit rate of the modulation scheme increases. Such switching of the voltage difference between the first input voltage and the second input voltage may be performed at the time of switching of a frame of the radio frequency signal. The pre-regulator circuit **10A** may be included in the tracker circuit **1B**, like the pre-regulator circuit **10** according to the modification of Embodiment 1.

[0206] According to an exemplary aspect, the digital control circuit **60B** is configured to control the pre-regulator circuit **10C** in addition to the switched-capacitor circuit **20** and the supply modulator **30** based on digital control signals from the RFIC **5**. Specifically, the digital control circuit **60B** is further configured to generate and output a control signal for controlling a switch included in the pre-regulator circuit **10C**. The circuit configuration of the digital control circuit **60B** will be described later with reference to FIG. **11**. It is also noted that the digital control circuit **60B** may be omitted from the tracker circuit **1B** in an exemplary aspect.

[3.2 Circuit Configuration of Tracker Circuit **1B**]

[0207] The circuit configuration of the tracker circuit **1B** will be described with reference to FIG. **11**. FIG. **11** is a circuit configuration diagram of the tracker circuit **1B** and the pre-regulator circuit **10A** according to this embodiment.

[0208] FIG. **11** illustrates an exemplary circuit configuration, and the tracker circuit **1B** and the pre-regulator circuit **10A** may be implemented by using any of a wide variety of circuit implementation and circuit techniques. Accordingly, the description of the individual circuits provided below should not be construed in a limiting manner.

[0209] As described above, the tracker circuit **1B** includes the pre-regulator circuit **10C**, the switched-capacitor circuit **20**, the supply modulator **30**, and the digital control circuit **60B**. The tracker circuit **1B** may include a filter circuit (not illustrated) between the supply modulator **30** and the power amplifier **2**.

[0210] Hereinafter, the circuit configurations of the pre-regulator circuit **10C** and the digital control circuit **60B** will be described in order.

[3.2.1 Circuit Configuration of Pre-Regulator Circuit **10C**]

[0211] First, the circuit configuration of the pre-regulator circuit **10C** will be described with reference to FIG. **11**. The pre-regulator circuit **10C** includes an input terminal **110C**, an output terminal **112C**, switches **S71C** and **S72C**, a power inductor **L71C**, and a capacitor **C72C**.

[0212] The input terminal **110C** is a terminal for receiving a first input voltage from the pre-regulator circuit **10A**. The input terminal **110C** is connected to the pre-regulator circuit **10A** outside the pre-regulator circuit **10C** and is connected to the switch **S71C** inside the pre-regulator circuit **10C**.

[0213] The output terminal **112C** is a terminal for supplying a second input voltage (**V1**) to the switched-capacitor circuit **20**. The output terminal **112C** is connected to the input terminal **122** of

the switched-capacitor circuit **20** outside the pre-regulator circuit **10C** and is connected to the power inductor **L71C** inside the pre-regulator circuit **10C**.

[0214] The power inductor **L71C** is an inductor used to raise and lower a direct current voltage. One end of the power inductor **L71C** is connected to the switches **S71C** and **S72C**, and the other end of the power inductor **L71C** is connected to the output terminal **112C**.

[0215] The switch **S71C** is connected between the input terminal **110C** and the one end of the power inductor **L71C**. Specifically, the switch **S71C** includes a terminal connected to the input terminal **110C** and a terminal connected to the one end of the power inductor **L71C**. In this connection configuration, ON/OFF switching of the switch **S71C** enables switching between connection and disconnection between the input terminal **110C** and the one end of the power inductor **L71C**.

[0216] The switch **S72C** is connected between the one end of the power inductor **L71C** and ground. Specifically, the switch **S72C** includes a terminal connected to the one end of the power inductor **L71C** and a terminal connected to ground. In this connection configuration, ON/OFF switching of the switch **S72C** enables switching between connection and disconnection between the one end of the power inductor **L71C** and ground.

[0217] The capacitor **C72C** is connected between ground and a path between the other end of the power inductor **L71C** and the output terminal **112C**. Specifically, one of the two electrodes of the capacitor **C72C** is connected to the other end of the power inductor **L71C** and the output terminal **112C**, and the other of the two electrodes of the capacitor **C72C** is connected to ground.

[0218] According to an exemplary aspect, the pre-regulator circuit **10C** having the above-described configuration is configured to lower a variable voltage **V3** (e.g., the first input voltage of the switched-capacitor circuit **20**) to a variable voltage **V1** (the second input voltage of the switched-capacitor circuit **20**).

[0219] The configuration of the pre-regulator circuit **10C** illustrated in FIG. **11** is illustrative and is not restrictive. For example, the switch **S72C** may be replaced with a diode. The pre-regulator circuit **10C** may be a boost converter. In this case, the pre-regulator circuit **10C** may be configured to receive a second input voltage from the pre-regulator circuit **10A** and raise the second input voltage to a first input voltage.

[3.2.2 Circuit Configuration of Digital Control Circuit **60B**]

[0220] The circuit configuration of the digital control circuit **60B** will then be described with reference to FIG. **11**. The digital control circuit **60B** includes a first controller **61B**, the second controller **62**, and the control terminals **161** to **164**.

[0221] According to an exemplary aspect, the first controller **61B** is configured to process source-synchronous digital control signals received from the RFIC **5** via the control terminals **161** and **162** and thereby generate control signals for controlling the switched-capacitor circuit **20** and the pre-regulator circuit **10C**. Turning ON/OFF of the switches **S71C** and **S72C** included in the pre-regulator circuit **10C** is controlled by a control signal from the first controller **61B**.

[0222] The digital control signals processed by the first controller **61B** are not limited to source-synchronous digital control signals. For example, the first controller **61B** may process clock-embedded digital control signals. The first controller **61B** may generate a control signal for controlling the supply modulator **30**.

[0223] Although one set of a clock signal and a data signal is used in this embodiment, the exemplary aspects of the present disclosure are not limited thereto. For example, a plurality of sets of a clock signal and a data signal may be used as digital control signals. For example, sets of a clock signal and a data signal may be individually used for the pre-regulator circuit **10C** and the switched-capacitor circuit **20**.

[3.3 Technical Effects and the Like]

[0224] As described above, the tracker circuit **1B** according to this embodiment further includes the pre-regulator circuit **10C** configured to lower the first input voltage to the second input voltage.

[0225] Accordingly, a one-input one-output buck-boost converter, which is often used in communication devices of the related art, may be used, and furthermore, the tracker circuit **1B** may be operated by one buck-boost converter. The tracker circuit **1B** may thus be introduced into the communication device **7B** more easily.

Additional Exemplary Embodiments

[0226] The tracker circuit, the communication device, and the tracking method according to the exemplary aspects of the present disclosure have heretofore been described based on the embodiments. However, the tracker circuit, the communication device, and the tracking method described herein are not limited to the embodiments above. Instead, an additional embodiment may be implemented by combining any components in the embodiments above, modifications obtained by applying various changes conceived by those skilled in the art to the embodiments above without departing from the gist of the exemplary aspects of the present disclosure, and various devices including the above-described tracker circuit are also included in the present disclosure.

[0227] For example, in the circuit configurations of the various circuits according to the embodiments above, another circuit element, wiring line, and the like may be inserted between individual circuit elements and paths connecting signal paths disclosed in the drawings. For example, an inductor and/or a capacitor may be inserted between the tracker circuit **1** and the power amplifier **2**.

[0228] In the embodiments above, a first input voltage is supplied to the node **N3** of the switched-capacitor circuit **20**, and a second input voltage is supplied to the node **N1**. However, the exemplary aspects of the present disclosure are not limited thereto. For example, the first input voltage may be supplied to the node **N2**, and the second input voltage may be supplied to the node **N1**. In this case, a third output voltage based on the voltage difference between the first input voltage and the second input voltage is output from the output terminal **123**. Alternatively, the first input voltage may be supplied to the node **N3**, and the second input voltage may be supplied to the node **N2**. In this case, a third output voltage based on the voltage difference between the first input voltage and the second input voltage is output from the output terminal **125**.

[0229] In the embodiments above, the switched-capacitor circuit **20** generates three discrete voltages; however, the number of discrete voltages is not limited to three. For example, the switched-capacitor circuit **20** may generate four or more discrete voltages.

[0230] The tracker circuit **1** and/or the tracker circuit **1B** according to the embodiments above may include a plurality of supply modulators. In this case, the tracker circuit **1** and/or the tracker circuit **1B** may supply different voltages to a plurality of power amplifiers.

[0231] The exemplary aspects of the present disclosure are widely usable, as a tracker circuit for supplying a voltage to a power amplifier, for communication devices such as mobile phones.

REFERENCE SIGNS LIST

[0232] **1**, **1B** tracker circuit [0233] **2** power amplifier [0234] **3** filter [0235] **4** radio frequency circuit [0236] **5** RFIC [0237] **6** antenna [0238] **7**, **7A**, **7B** communication device [0239] **10**, **10A**, **10B**, **10C** pre-regulator circuit [0240] **20** switched-capacitor circuit [0241] **30** supply modulator [0242] **50** direct current power source [0243] **60**, **60B** digital control circuit [0244] **61**, **61B** first controller [0245] **62** second controller [0246] **110**, **110A**, **110B**, **110C**, **121**, **122**, **131**, **132**, **133** input terminal [0247] **111**, **111A**, **112**, **112B**, **112C**, **123**, **124**, **125**, **134** output terminal [0248] **161**, **162**, **163**, **164** control terminal

Claims

1. A tracker circuit comprising: a switched-capacitor circuit configured to generate a plurality of discrete voltages; and a supply modulator configured to selectively output at least one discrete voltage of the plurality of discrete voltages to a power amplifier, wherein the switched-capacitor circuit is configured to generate the plurality of discrete voltages based on a voltage difference

between a first input voltage and a second input voltage that is variable and that is lower than the first input voltage.

2. The tracker circuit according to claim 1, wherein the first input voltage is variable.

3. The tracker circuit according to claim 1, wherein the plurality of discrete voltages include a first output voltage based on the first input voltage, a second output voltage based on the second input voltage, and a third output voltage based on a voltage difference between the first input voltage and the second input voltage.

4. The tracker circuit according to claim 3, wherein the third output voltage is lower than the first output voltage.

5. The tracker circuit according to claim 3, wherein the switched-capacitor circuit is configured to generate the third output voltage to cause each of a voltage difference between the first output voltage and the third output voltage and a voltage difference between the second output voltage and the third output voltage to increase in response to an increase in the voltage difference between the first input voltage and the second input voltage.

6. The tracker circuit according to claim 1, wherein the voltage difference between the first input voltage and the second input voltage is switched in accordance with a modulation scheme for a radio frequency signal amplified by the power amplifier.

7. The tracker circuit according to claim 6, wherein the voltage difference between the first input voltage and the second input voltage increases as a bit rate of the modulation scheme for increases of the radio frequency signal.

8. The tracker circuit according to claim 6, wherein the voltage difference between the first input voltage and the second input voltage is switched at a time of switching of a frame of the radio frequency signal amplified by the power amplifier.

9. The tracker circuit according to claim 1, wherein the tracker circuit is connected to a converter circuit configured to convert a power supply voltage into the first input voltage and the second input voltage.

10. The tracker circuit according to claim 1, wherein: the tracker circuit is connected to a first converter circuit and a second converter circuit, the first converter circuit is configured to convert a power supply voltage into the first input voltage, and the second converter circuit is configured to convert the power supply voltage into the second input voltage.

11. The tracker circuit according to claim 1, further comprising a converter circuit configured to lower the first input voltage to the second input voltage.

12. A communication device comprising: the tracker circuit according to claim 1; a converter circuit configured to convert a power supply voltage into the first input voltage and the second input voltage; a signal processing circuit configured to process a radio frequency signal; and a radio frequency circuit including the power amplifier and configured to transmit the radio frequency signal from between the signal processing circuit and an antenna.

13. A tracker circuit comprising: a switched-capacitor circuit that includes a first input terminal and a second input terminal and that is configured to generate a plurality of discrete voltages, the first input terminal configured to receive a first input voltage, the second input terminal configured to receive a second input voltage that is variable and that is lower than the first input voltage; a supply modulator configured to selectively output at least one discrete voltage of the plurality of discrete voltages to a power amplifier; a first capacitor connected between the first input terminal and ground; and a second capacitor connected between the second input terminal and ground.

14. The tracker circuit according to claim 13, further comprising a converter circuit that includes: a power inductor, a first output terminal connected to the first input terminal of the switched-capacitor circuit, a second output terminal connected to the second input terminal of the switched-capacitor circuit, a first switch connected between an output end of the power inductor and the first output terminal, a second switch connected between the output end of the power inductor and the second output terminal, a third switch connected between the output end of the power inductor and

ground, and the first capacitor and the second capacitor, wherein the first capacitor is connected between ground and a path between the first switch and the first output terminal, and wherein the second capacitor is connected between ground and a path between the second switch and the second output terminal.

15. The tracker circuit according to claim 14, wherein the switched-capacitor circuit includes a switch having a withstand voltage that is higher than a withstand voltage of each of the first switch and the second switch included in the converter circuit.

16. The tracker circuit according to claim 13, wherein the plurality of discrete voltages include a first output voltage based on the first input voltage, a second output voltage based on the second input voltage, and a third output voltage based on a voltage difference between the first input voltage and the second input voltage.

17. The tracker circuit according to claim 16, wherein the third output voltage is lower than the first output voltage.

18. The tracker circuit according to claim 16, wherein the switched-capacitor circuit is configured to generate the third output voltage to cause each of a voltage difference between the first output voltage and the third output voltage and a voltage difference between the second output voltage and the third output voltage to increase in response to an increase in the voltage difference between the first input voltage and the second input voltage.

19. A communication device comprising: the tracker circuit according to claim 13; a converter circuit configured to convert a power supply voltage into the first input voltage and the second input voltage; a signal processing circuit configured to process a radio frequency signal; and a radio frequency circuit including the power amplifier and configured to transmit the radio frequency signal from between the signal processing circuit and an antenna.

20. A tracking method comprising: receiving a first input voltage and a second input voltage lower than the first input voltage; generating a plurality of first discrete voltages based on a first voltage difference between the first input voltage and the second input voltage; selectively outputting at least one discrete voltage of the plurality of first discrete voltages to a power amplifier; receiving a third input voltage and a fourth input voltage that is lower than the third input voltage, the fourth input voltage being a different voltage than the second input voltage; generating a plurality of second discrete voltages based on a second voltage difference between the third input voltage and the fourth input voltage; and selectively outputting at least one discrete voltage of the plurality of second discrete voltages to the power amplifier.
